Bearing Tolerances

The respective dimensional accuracy, profile accuracy and running accuracy of the bearing bore diameter (d), cylindrical roller outer diameter (D), and outer ring width (C) are as shown in **Table 4.3** of Section 4. "**Bearing Accuracy**" (page A-26). (Conforming to JIS Accuracy Class-0)

On the other hand, the dimensional tolerances for spherical roller outer diameter (D) and the roller inscribed circle diameter (F_W) of **Type RNA22** are as shown in applicable Dimensions Table.

Bearing fits and radial clearance

The tolerance range class of shaft against bearing with inner ring shall be g6 (h6) and, where a shaft is used as the direct raceway surface (**Type RNA22**), the tolerance range class of the shaft shall be k5 (k6). In general, the outer ring is not fitted in a housing. Mounting relations

Table 1 Radial clearances

Unit: μm

Nominal roller inscribed circle dia. F_W (mm)		Clearance							
		C2		Ordinary		C3		C4	
over	Incl.	min	max	min	max	min	max	min	max
3	6	0	10	3	17	15	30	20	40
6	10	0	12	5	20	15	30	25	45
10	18	0	15	5	25	15	35	30	55
18	30	0	20	10	30	20	40	40	65
30	50	0	25	10	40	25	55	50	80
50	80	0	30	15	50	30	65	60	100
80	100	0	35	20	55	35	75	70	115

(1) The side face height in the roller follower mount must be made larger than "e" dimension described in applicable Dimensions Table. (**Fig.1**) In mounting, chamfer the mounting surface at R as small as possible (around 0.5×45°) and bring the inner ring and the end face of side plate in precise contact with one another.

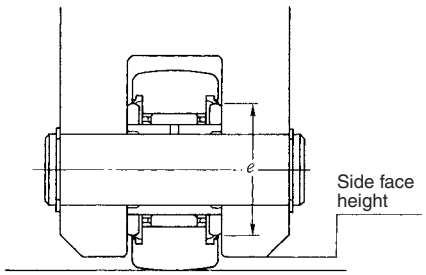


Fig.1

(2) Where the roller follower is mounted, locate the inner ring oil hole within the non-load area (load free side). (**Fig. 2**)
If the oil hole locates within the load area, it would cause shorter life.

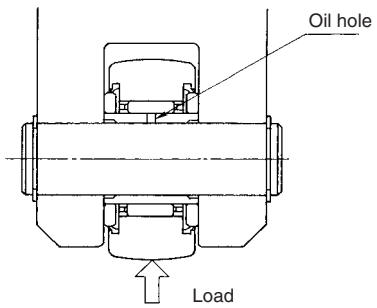


Fig.2

(3) **Type NA··22LL** and **RNA22** are of separable type and, hence, the outer ring is guided by flange or thrust washer mounted on shaft (pin). Therefore, the guide surface must be finished more precisely than by lathe-turning and deburred completely for surface smoothing. In addition, when the guide surface is not hardened the outer ring must be guided at A-dimension shown in **Fig.3**. When it is hardened, even a little smaller guide surface can be used.

$$A \geqq \frac{1}{2}(D+e)$$

For D and e dimensions refer to applicable Dimensions Table.

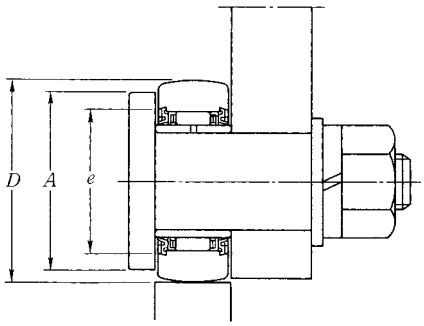


Fig.3

In assembling this bearing, it must be handled with good care to protect the seal lip from bending and flaw.

Where any of NTN roller followers was stub-mounted, non-uniform load (bias load) could act on the bearing, inversely affected by fitting loose arising from further continued running.
Good care must be exercised of such fitting loose, for stable running of the equipment.

Lubrication

The roller followers with synthetic rubber seal (with tail code : **LL**) and the full complement roller type followers are filled up with lithium soap base grease so that they can be used in the temperature range of -25 to +100˚C. Lubricant feeding and replenishing into the bearing are done through an oil hole provided on the inner ring. In this case, the inner ring must be fitted so the oil hole locates within non-load area. For the full complement roller type roller followers **NATV**, **NACV**, **NUTR** and **NUTW** with no cage, it is necessary to shorten the grease replenishing interval.

Further, a roller follower with cage and without seal is not filled up with lubrication grease. When needing a follower with grease-filled cage, feel free to contact **NTN**.

The outer ring outer surface of bearing and the track surface must both be lubricated. Failure to lubricate could result earlier in damage of the bearing.

Track load capacity

Statically allowable maximum radial load on the contact surface of track to track roller is referred to as "track load capacity", which differs depending on the track hardness. The track load capacity described in applicable Dimensions Table is subject to track hardness HRC40. If the hardness value is other than HRC40, multiply the track load capacity in the Dimensions Table by the adjustment factor G in **Table 2**, for determining the track load capacity. However, the track load capacity shall be C_{0r} if the calculated track load capacity exceeds the basic static load rating C_{0r} of bearing.

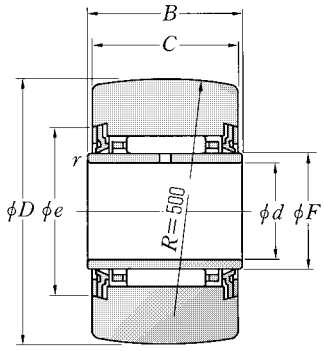
Refer to page A-55 for the detail.

Table 2 Adjustment factor G

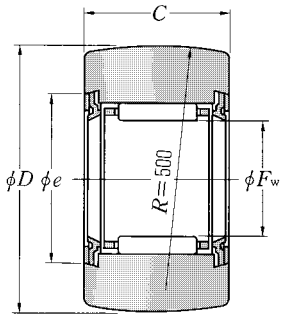
Hardness HRC	Adjustment factor G	
	Cylindrical surface	Spherical surface
20	0.37	0.22
25	0.46	0.31
30	0.58	0.45
35	0.75	0.65
40	1.00	1.00
45	1.40	1.70
50	2.00	2.80
55	2.80	4.70

Metric series

Type NA22··LL
Type RNA22··LL



Type NA22··LL
(With inner ring)



Type RNA22··LL
(Without inner ring)

D 19～90mm

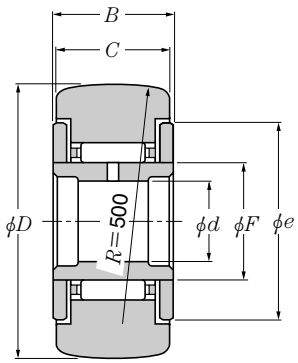
Boundary dimentions mm								Basic load ratings N dynamic	Basic load ratings N static	Basic load ratings kgf dynamic	Basic load ratings kgf static	Load ratings of truck N		Load ratings of truck kgf	
D $^{0}_{-0.05}$	B	C	d	F_w	e	F	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}				
19	12	11.8	6	$10^{+0.022}_{+0.013}$	16	10	0.3	4 550	4 250	460	435	1 380	4 400	141	445
24	12	11.8	8	$12^{+0.027}_{+0.016}$	18	12	0.3	5 150	5 250	525	535	1 900	5 500	193	565
30	14	13.8	10	$14^{+0.027}_{+0.016}$	20	14	0.3	7 550	9 000	770	915	2 620	7 550	267	770
32	14	13.8	12	$16^{+0.027}_{+0.016}$	22	16	0.3	8 100	10 300	830	1 050	2 860	8 050	291	820
35	14	13.8	15	$20^{+0.033}_{+0.020}$	26	20	0.3	9 850	14 100	1 010	1 440	3 200	8 800	325	900
40	16	15.8	17	$22^{+0.033}_{+0.020}$	28	22	0.3	10 400	15 600	1 060	1 590	3 850	10 900	390	1 110
47	18	17.8	20	$25^{+0.033}_{+0.020}$	33	25	0.3	16 900	22 900	1 730	2 340	4 700	14 800	480	1 510
52	18	17.8	25	$30^{+0.033}_{+0.020}$	38	30	0.3	17 900	25 900	1 820	2 640	5 550	16 400	565	1 670
62	20	19.8	30	$35^{+0.041}_{+0.025}$	43	35	0.3	21 400	34 500	2 190	3 500	6 950	22 200	710	2 260
72	23	22.7	35	$42^{+0.041}_{+0.025}$	50	42	0.6	26 300	47 500	2 690	4 850	8 050	28 700	820	2 930
80	23	22.7	40	$48^{+0.041}_{+0.025}$	57	48	0.6	28 400	55 000	2 900	5 600	9 800	32 000	1 000	3 250
85	23	22.7	45	$52^{+0.049}_{+0.030}$	62	52	0.6	29 300	58 500	2 990	5 950	10 400	34 000	1 060	3 450
90	23	22.7	50	$58^{+0.049}_{+0.030}$	68	58	0.6	31 000	66 000	3 200	6 700	11 400	36 000	1 160	3 650

Note 1) Allowable minimum chamfer dimension r.
2) For bearing with cylindrical outer surface, the nominal bearing is followed by "**X**". In this case, the bearing is manufactured with outer ring diameter tolerance conforming to JIS Class-0. Ex. **NA2203XLL**

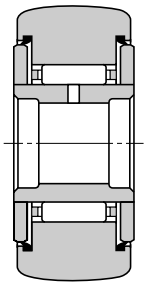
Limiting speeds r/min grease	Bearing numbers		Mass (approx.) kg	
	Type NA22··LL[2)]	Type RNA22··LL[2)]	Type NA22··LL	Type RNA22··LL
10 000	**NA22/6LL**	**RNA22/6LL**	0.023	0.018
10 000	**NA22/8LL**	**RNA22/8LL**	0.035	0.027
10 000	**NA2200LL**	**RNA2200LL**	0.060	0.052
9 500	**NA2201LL**	**RNA2201LL**	0.067	0.057
7 500	**NA2202LL**	**RNA2202LL**	0.075	0.060
7 000	**NA2203LL**	**RNA2203LL**	0.113	0.094
6 000	**NA2204LL**	**RNA2204LL**	0.176	0.152
5 000	**NA2205LL**	**RNA2205LL**	0.209	0.179
4 300	**NA2206LL**	**RNA2206LL**	0.322	0.284
3 600	**NA2207LL**	**RNA2207LL**	0.506	0.432
3 100	**NA2208LL**	**RNA2208LL**	0.623	0.530
2 900	**NA2209LL**	**RNA2209LL**	0.638	0.545
2 600	**NA2210LL**	**RNA2210LL**	0.682	0.563

Metric series

Type NATR
Type NATR··LL
Type NATV
Type NATV··LL



Type NATR
(With cage)



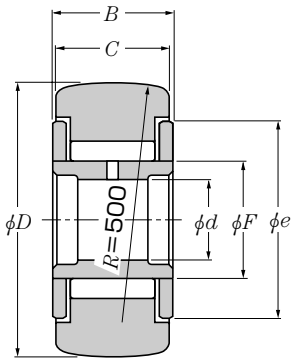
Type NATR··LL
(Sealed type with cage)

D 16～90mm

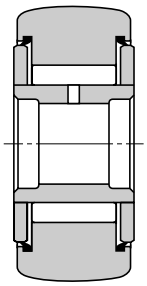
Boundary dimensions mm						Basic load ratings N		Basic load ratings kgf		Load ratings of truck N		Load ratings of truck kgf		Limiting speeds r/min	
						dynamic	static	dynamic	static						
D $^{0}_{-0.05}$	*B*	*C*	*d*	*e*	F_w	C_r	C_{or}	C_r	C_{or}					grease	oil
16	12 $^{0}_{-0.180}$	11	5	12	8	4 050	4 200	415	430	1 080	3 400	110	350	※19 000	※25 000
	12 $^{-0.180}$	11	5	12	8	6 500	9 350	665	955	1 080	3 400	110	350	※13 000	※16 000
19	12 $^{0}_{-0.180}$	11	6	14	10	4 750	5 400	480	555	1 380	4 050	141	415	※15 000	※20 000
	12 $^{-0.180}$	11	6	14	10	7 450	11 700	760	1 190	1 380	4 050	141	415	10 000	※13 000
24	15 $^{0}_{-0.180}$	14	8	19	12	6 900	7 700	705	785	1 900	6 650	193	680	※12 000	※16 000
	15 $^{-0.180}$	14	8	19	12	10 700	16 200	1 090	1 650	1 900	6 650	193	680	8 500	※11 000
30	15 $^{0}_{-0.180}$	14	10	23	15	7 850	9 650	800	985	2 620	7 700	267	785	10 000	※13 000
	15 $^{-0.180}$	14	10	23	15	12 000	20 300	1 230	2 070	2 620	7 700	267	785	6 500	8 500
32	15 $^{0}_{-0.180}$	14	12	25	17	8 850	11 700	900	1 190	2 860	8 200	291	835	9 000	※12 000
	15 $^{-0.180}$	14	12	25	17	13 000	23 000	1 330	2 350	2 860	8 200	291	835	6 000	7 500
35	19 $^{0}_{-0.210}$	18	15	27	20	13 300	20 800	1 360	2 120	3 200	11 900	325	1 220	7 500	10 000
	19 $^{-0.210}$	18	15	27	20	18 400	38 000	1 870	3 900	3 200	11 900	325	1 220	5 000	6 500
40	21 $^{0}_{-0.210}$	20	17	32	22	14 000	22 800	1 430	2 330	3 850	14 500	390	1 480	7 000	9 000
	21 $^{-0.210}$	20	17	32	22	19 400	42 000	1 980	4 250	3 850	14 500	390	1 480	4 500	6 000
47	25 $^{0}_{-0.210}$	24	20	37	25	20 700	33 500	2 110	3 450	4 700	21 000	480	2 150	6 000	8 000
	25 $^{-0.210}$	24	20	37	25	28 800	61 000	2 940	6 250	4 700	21 000	480	2 150	4 000	5 000
52	25 $^{0}_{-0.210}$	24	25	42	30	22 800	40 500	2 320	4 100	5 500	23 300	565	2 370	5 000	6 500
	25 $^{-0.210}$	24	25	42	30	31 500	73 500	3 200	7 500	5 500	23 300	565	2 370	3 300	4 500
62	29 $^{0}_{-0.210}$	28	30	51	38	36 000	66 000	3 650	6 750	6 950	33 000	710	3 350	4 000	5 500
	29 $^{-0.210}$	28	30	51	38	47 500	115 000	4 850	11 700	6 950	33 000	710	3 350	2 600	3 500
72	29 $^{0}_{-0.210}$	28	35	58	44.5	39 000	77 000	3 950	7 850	8 050	37 000	820	3 750	3 300	4 500
	29 $^{-0.210}$	28	35	58	44.5	52 000	134 000	5 300	13 600	8 050	37 000	820	3 750	2 200	2 900
80	32 $^{0}_{-0.250}$	30	40	66	50	49 500	92 500	5 050	9 400	9 800	44 500	1 000	4 500	3 000	4 000
	32 $^{-0.250}$	30	40	66	50	68 500	171 000	7 000	17 500	9 800	44 500	1 000	4 500	2 000	2 600
85	32 $^{0}_{-0.250}$	30	45	71	55	51 500	100 000	5 250	10 200	10 400	47 000	1 060	4 800	2 700	3 600
90	32 $^{0}_{-0.250}$	30	50	76	60	53 000	108 000	5 450	11 000	11 400	50 000	1 160	5 100	2 500	3 300
	32 $^{-0.250}$	30	50	76	60	76 000	205 000	7 750	20 900	11 400	50 000	1 160	5 100	1 600	2 100

Note 1) For bearing with cylindrical outer surface, the nominal bearing is followed by "**X**". In this case, the bearing is manufactured with outer ring diameter tolerance conforming to JIS Class-0. Ex. **NATR5X**

Remarks: For the follower with seal, the maximum revolutions is allowable up to 10 000 r/min.



Type NATV
(Full complement type)

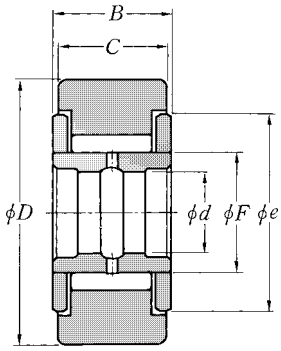


Type NATV··LL
(Full complement type with seals)

Bearing numbers [1]		Mass
Type NATR Type NATV	Type NATR··LL Type NATV··LL	kg (approx.)
NATR5	NATR5LL	0.018
NATV5	NATV5LL	0.020
NATR6	NATR6LL	0.025
NATV6	NATV6LL	0.027
NATR8	NATR8LL	0.042
NATV8	NATV8LL	0.044
NATR10	NATR10LL	0.061
NATV10	NATV10LL	0.065
NATR12	NATR12LL	0.069
NATV12	NATV12LL	0.074
NATR15	NATR15LL	0.098
NATV15	NATV15LL	0.102
NATR17	NATR17LL	0.140
NATV17	NATV17LL	0.145
NATR20	NATR20LL	0.246
NATV20	NATV20LL	0.254
NATR25	NATR25LL	0.275
NATV25	NATV25LL	0.285
NATR30	NATR30LL	0.470
NATV30	NATV30LL	0.481
NATR35	NATR35LL	0.635
NATV35	NATV35LL	0.647
NATR40	NATR40LL	0.875
NATV40	NATV40LL	0.890
NATR45	NATR45LL	0.910
NATR50	NATR50LL	0.960
NATV50	NATV50LL	0.990

Inch series

Type NACV··X◦
Type NACV··XLL

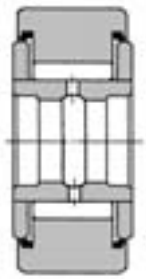


Type NACV··X
(Full complement)

D 19.050～152.400mm

Boundary dimentions mm (1/25.4mm)						Basic load ratings N		kgf	
						dynamic	static	dynamic	static
D $^{0}_{-0.025}$	B $^{0}_{-0.250}$	C $^{0}_{-0.130}$	d	e	F_w	C_r	C_{or}	C_r	C_{or}
19.050(3/4)	14.288(9/16)	12.700(1/2)	6.350(1/4) $^{+0.005}_{-0.010}$	15.5	11	8 050	13 300	825	1 360
22.225(7/8)	14.288(9/16)	12.700(1/2)	6.350(1/4) $^{+0.005}_{-0.010}$	15.5	11	8 050	13 300	825	1 360
25.400(1)	17.462(11/16)	15.875(5/8)	7.938(5/16) $^{+0.005}_{-0.010}$	19.5	14	11 700	18 900	1 190	1 920
28.575(1 1/8)	17.462(11/16)	15.875(5/8)	7.938(5/16) $^{+0.005}_{-0.010}$	19.5	14	11 700	18 900	1 190	1 920
31.750(1 1/4)	20.638(13/16)	19.050(3/4)	9.525(3/8) $^{+0.005}_{-0.010}$	25	18.47	17 700	35 000	1 810	3 600
34.925(1 3/8)	20.638(13/16)	19.050(3/4)	9.525(3/8) $^{+0.005}_{-0.010}$	25	18.47	17 700	35 000	1 810	3 600
38.100(1 1/2)	23.812(15/16)	22.225(7/8)	11.112(7/16) $^{+0.005}_{-0.010}$	27	21	21 100	45 500	2 150	4 650
41.275(1 5/8)	23.812(15/16)	22.225(7/8)	11.112(7/16) $^{+0.005}_{-0.010}$	27	21	21 100	45 500	2 150	4 650
44.450(1 3/4)	26.988(1 1/16)	25.400(1)	12.700(1/2) $^{+0.005}_{-0.010}$	31.5	24.65	28 400	60 500	2 900	6 150
47.625(1 7/8)	26.988(1 1/16)	25.400(1)	12.700(1/2) $^{+0.005}_{-0.010}$	31.5	24.65	28 400	60 500	2 900	6 150
50.800(2)	33.338(1 5/16)	31.750(1 1/4)	15.875(5/8) $^{+0.005}_{-0.010}$	36.5	26.71	41 000	87 500	4 200	8 950
57.150(2 1/4)	33.338(1 5/16)	31.750(1 1/4)	15.875(5/8) $^{+0.005}_{-0.010}$	36.5	26.71	41 000	87 500	4 200	8 950
63.500(2 1/2)	39.688(1 9/16)	38.100(1 1/2)	19.050(3/4) $^{+0.005}_{-0.010}$	44	31.15	54 500	119 000	5 600	12 200
69.850(2 3/4)	39.688(1 9/16)	38.100(1 1/2)	19.050(3/4) $^{+0.005}_{-0.010}$	44	31.15	54 500	119 000	5 600	12 200
76.200(3)	46.038(1 13/16)	44.450(1 3/4)	25.400(1) $^{+0.002}_{-0.013}$	53	36.85	76 500	177 000	7 800	18 000
82.550(3 1/4)	46.038(1 13/16)	44.450(1 3/4)	25.400(1) $^{+0.002}_{-0.013}$	53	36.85	76 500	177 000	7 800	18 000
88.900(3 1/2)	52.388(2 1/16)	50.800(2)	28.575(1 1/8) $^{+0.002}_{-0.013}$	60	44.5	84 500	214 000	8 650	21 800
101.600(4)	58.738(2 5/16)	57.150(2 1/4)	31.750(1 1/4) $^{+0.002}_{-0.013}$	63	44.5	106 000	244 000	10 800	24 900
127.000(5)	73.025(2 7/8)	69.850(2 3/4)	44.450(1 3/4) $^{+0.002}_{-0.013}$	89	68.7	189 000	520 000	19 300	53 000
152.400(6)	85.725(3 3/8)	82.550(3 1/4)	57.150(2 1/4) $^{+0.002}_{-0.013}$	110	81.35	260 000	675 000	26 500	68 500

Note 1) For the bearing with spherical outer surface, "**X**" symbol shall be deleted from the Nominal number. In this case, spherical bearing (R=500) is manufactured with tolerance 0/ -0.05 against the outer ring outer diameter "D". Ex. **NACV12**



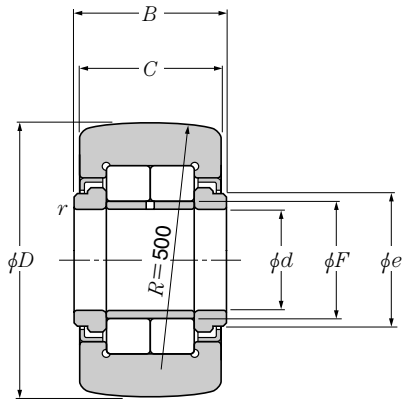
Type NACV··XLL
(Full complement sealed type)

Load ratings of truck				Limiting speeds r/min		Bearing numbers		Mass (approx.) kg
N		kgf		grease	oil	Type NACV··X[4)]	Type NACV··XLL[1)]	Type NACV··X
4 600	1 380	470	140	9 000	※11 000	**NACV12X**	**NACV12XLL**	0.027
5 350	1 710	545	174	9 000	※11 000	**NACV14X**	**NACV14XLL**	0.036
7 400	2 060	755	210	7 100	9 200	**NACV16X**	**NACV16XLL**	0.059
8 350	2 430	850	248	7 100	9 200	**NACV18X**	**NACV18XLL**	0.073
11 400	2 840	1 160	290	5 400	7 000	**NACV20X**	**NACV20XLL**	0.109
12 500	3 250	1 280	330	5 400	7 000	**NACV22X**	**NACV22XLL**	0.132
16 300	3 600	1 660	365	4 800	6 200	**NACV24X**	**NACV24XLL**	0.177
17 600	4 050	1 800	410	4 800	6 200	**NACV26X**	**NACV26XLL**	0.218
21 600	4 400	2 200	450	4 100	5 300	**NACV28X**	**NACV28XLL**	0.281
23 200	4 850	2 360	495	4 100	5 300	**NACV30X**	**NACV30XLL**	0.327
31 000	5 300	3 150	540	3 700	4 800	**NACV32X**	**NACV32XLL**	0.454
35 000	6 200	3 550	635	3 700	4 800	**NACV36X**	**NACV36XLL**	0.585
44 500	7 200	4 550	735	3 200	4 100	**NACV40X**	**NACV40XLL**	0.902
49 000	8 250	5 000	840	3 200	4 100	**NACV44X**	**NACV44XLL**	1.05
64 000	9 150	6 500	935	2 700	3 500	**NACV48X**	**NACV48XLL**	1.39
69 000	10 000	7 050	1 020	2 700	3 500	**NACV52X**	**NACV52XLL**	1.66
86 000	11 100	8 800	1 130	2 200	2 800	**NACV56X**	**NACV56XLL**	2.19
112 500	13 200	11 500	1 350	2 200	2 800	**NACV64X**	**NACV64XLL**	3.22
165 500	17 900	16 900	1 830	1 500	1 900	**NACV80X**	**NACV80XLL**	6.08
239 500	22 100	24 400	2 250	1 200	1 500	**NACV96X**	**NACV96XLL**	10.0

Remarks: For the follower with seal, the maximum revolutions is allowable up to 10 000 r/min.

Metric series

Type NUTR2
Type NUTR3



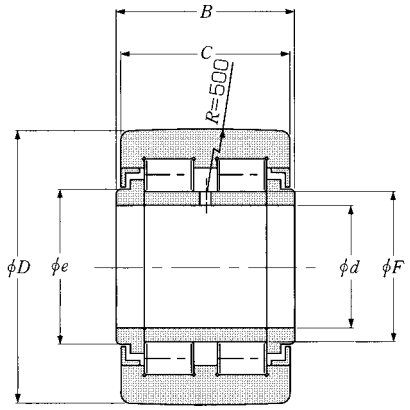
D 35～110mm

Boundary dimensions mm							Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Load ratings of truck N		kgf		Limiting speeds r/min grease	Bearing numbers [2]	Mass kg
D $^{0}_{-0.05}$	B	C	d	e	F_w	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}							(approx.)
35	19 $^{0}_{-0.210}$	18	15	20	19	0.3	22 300	25 700	2 280	2 620	3 200	11 900	325	1 220	5 500	**NUTR202**	0.100
40	21 $^{0}_{-0.210}$	20	17	22	21.5	0.3	24 100	29 100	2 450	2 970	3 850	14 500	390	1 480	4 700	**NUTR203**	0.147
42	19 $^{0}_{-0.210}$	18	15	20	19	0.3	22 300	25 700	2 280	2 620	4 100	14 300	415	1 460	5 500	**NUTR302**	0.160
47	21 $^{0}_{-0.210}$	20	17	22	21.5	0.3	24 100	29 100	2 450	2 970	4 700	17 000	480	1 740	4 700	**NUTR303**	0.222
	25 $^{0}_{-0.210}$	24	20	27	25.5	0.3	38 500	48 000	3 950	4 900	4 700	21 000	480	2 150	4 000	**NUTR204**	0.245
52	25 $^{0}_{-0.210}$	24	20	27	25.5	0.3	38 500	48 000	3 950	4 900	5 550	23 300	565	2 370	4 000	**NUTR304**	0.321
	25 $^{0}_{-0.210}$	24	25	31	30	0.3	42 500	57 500	4 350	5 850	5 550	23 300	565	2 370	3 300	**NUTR205**	0.281
62	25 $^{0}_{-0.210}$	24	25	31	30	0.3	42 500	57 500	4 350	5 850	6 950	27 800	710	2 830	3 300	**NUTR305**	0.450
	29 $^{0}_{-0.210}$	28	30	38	35	0.3	56 500	72 500	5 750	7 400	6 950	33 000	710	3 350	2 900	**NUTR206**	0.466
72	29 $^{0}_{-0.210}$	28	30	38	35	0.3	56 500	72 500	5 750	7 400	8 050	38 500	820	3 900	2 900	**NUTR306**	0.697
	29 $^{0}_{-0.210}$	28	35	44	41.5	0.6	62 000	85 500	6 350	8 700	8 050	37 000	820	3 750	2 400	**NUTR207**	0.630
80	29 $^{0}_{-0.210}$	28	35	44	41.5	0.6	62 000	85 500	6 350	8 700	9 800	41 000	1 000	4 150	2 400	**NUTR307**	0.840
	32 $^{0}_{-0.250}$	30	40	51	47.5	0.6	87 000	125 000	8 850	12 700	9 800	44 500	1 000	4 500	2 100	**NUTR208**	0.817
85	32 $^{0}_{-0.250}$	30	45	55	52.5	0.6	92 000	137 000	9 350	14 000	10 400	47 000	1 060	4 800	1 900	**NUTR209**	0.883
90	32 $^{0}_{-0.250}$	30	40	51	47.5	0.6	87 000	125 000	8 850	12 700	11 400	50 000	1 160	5 100	2 100	**NUTR308**	1.13
	32 $^{0}_{-0.250}$	30	50	60	57	0.6	96 500	150 000	9 800	15 300	11 400	50 000	1 160	5 100	1 800	**NUTR210**	0.950
100	32 $^{0}_{-0.250}$	30	45	55	52.5	0.6	92 000	137 000	9 350	14 000	13 000	55 500	1 330	5 650	1 900	**NUTR309**	1.40
110	32 $^{0}_{-0.250}$	30	50	60	57	0.6	96 500	150 000	9 800	15 300	14 700	61 000	1 500	6 200	1 800	**NUTR310**	1.69

Note 1) Allowable minimum chamfer dimension r.
2) For bearing with cylindrical outer surface, the nominal bearing is followed by "**X**". In this case, the bearing is manufactured with outer ring diameter tolerance conforming to JIS Class-0. Ex. **NUTR208X**

Metric series

Type NUTW2



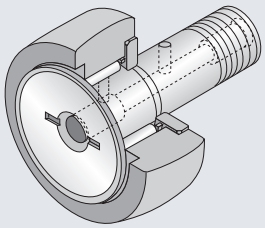
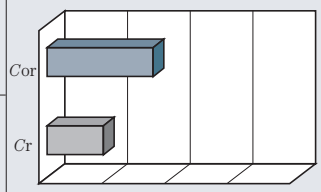
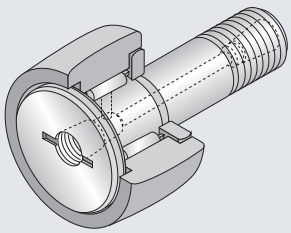
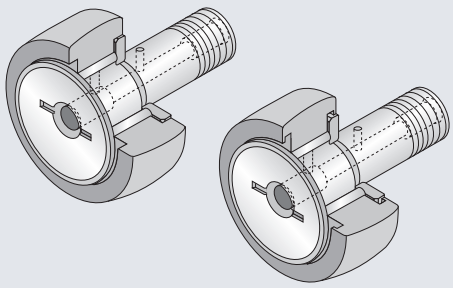
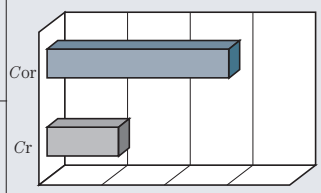
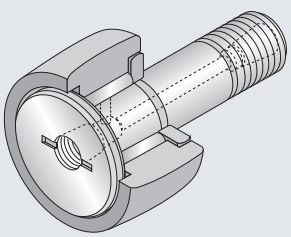
D 35～90mm

Boundary dimensions mm							Basic load ratings dynamic N		Basic load ratings static N		Load ratings of truck dynamic kgf		Load ratings of truck static kgf		Limiting speeds r/min grease	Bearing numbers [2]	Mass kg
D $^{0}_{-0.05}$	B	C	d	e	F_w	$r_{s\,min}$ [1]	C_r		C_{or}		C_r		C_{or}				(approx.)
35	22 $^{0}_{-0.210}$	21	15	20	19	0.3	24 100		28 300		2 460		2 880		5 500	**NUTW202**	0.115
40	24 $^{0}_{-0.210}$	23	17	22	21.5	0.3	26 000		32 000		2 650		3 250		4 700	**NUTW203**	0.167
47	29 $^{0}_{-0.210}$	28	20	27	25.5	0.3	40 500		51 500		4 150		5 250		4 000	**NUTW204**	0.280
52	29 $^{0}_{-0.210}$	28	25	31	30	0.3	45 000		61 500		4 600		6 250		3 300	**NUTW205**	0.322
62	35 $^{0}_{-0.210}$	34	30	38	35	0.3	59 500		77 000		6 050		7 900		2 900	**NUTW206**	0.549
72	35 $^{0}_{-0.210}$	34	35	44	41.5	0.6	65 000		91 000		6 650		9 250		2 400	**NUTW207**	0.747
80	38 $^{0}_{-0.250}$	36	40	51	47.5	0.6	90 500		131 000		9 250		13 400		2 100	**NUTW208**	0.953
85	38 $^{0}_{-0.250}$	36	45	55	52.5	0.6	95 500		144 000		9 750		14 700		1 900	**NUTW209**	1.03
90	38 $^{0}_{-0.250}$	36	50	60	57	0.6	100 000		158 000		10 200		16 100		1 800	**NUTW210**	1.11
											3 200	14 200	325	1 450			
											3 850	17 100	390	1 740			
											4 700	25 100	480	2 560			
											5 550	27 700	565	2 830			
											6 950	41 000	710	4 200			
											8 050	46 000	820	4 700			
											9 800	54 500	1 000	5 550			
											10 400	58 000	1 060	5 900			
											11 400	61 500	1 160	6 250			

Cam Followers (Stud Type Track Roller)

This is a track roller complete with stud instead of inner ring, which of the outer ring rolls on the track. These **NTN** stud type track rollers are applied to eccentric roller, guide roller, rocker arm roller, etc. similarly to the roller followers. One end of the stud is so threaded as to facilitate mounting thereof. And the outer ring is guided in axial direction by the side plates which are press-fitted in the flange end of the stud and the stud itself.

The outer ring has a thick wall and both of spherical outer profile and cylindrical outer surface (tail code: **X**) are available for it similarly to the outer ring of the roller follower.

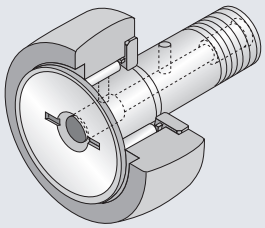
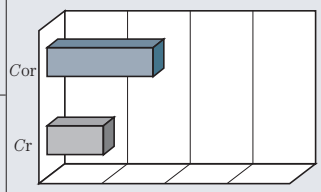
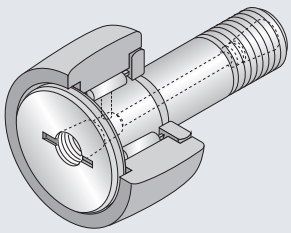
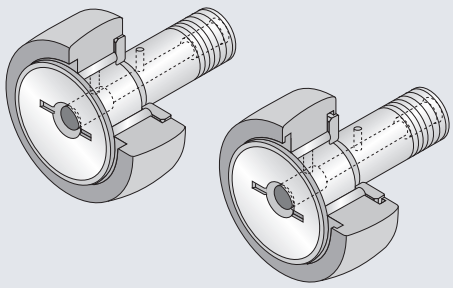
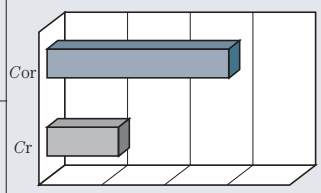
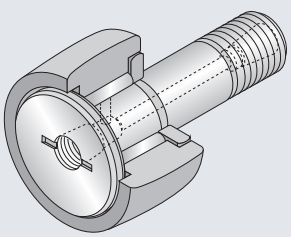
Follower type	Applied shaft diameter (mm)	Load capacity	Composition of nominal bearing number
KR **CR** 	KR : ϕ3～ϕ30 CR : ϕ4.826～ϕ22.225	 **KR90**	**KR 12 T2 H / 3A** Tailed code T2: Resin cage H: With hexagon hole 3A: grease Dimension code Type code
KRT 	ϕ6～ϕ30		**KRT 32 X LL** Tail code LL: Seal Tailed code X: Cylinder outer diameter Dimension code Type code
KRV **CRV** 	KRV : ϕ3～ϕ30 CRV : ϕ4.826～ϕ63.5	 **KRV90**	**CRV 30 X LL** Tail code LL: Seal Tailed code X: Cylinder outer diameter Dimension code Type code
KRVT 	ϕ6～ϕ30		**KRVT 52 X LL** Tail code LL: Seal Tailed code X: Cylinder outer diameter Dimension code Type code

※Each listed load capacity is subject to the reference bearing diameter of ϕ30.

Cam Followers (Stud Type Track Roller)

This is a track roller complete with stud instead of inner ring, which of the outer ring rolls on the track. These **NTN** stud type track rollers are applied to eccentric roller, guide roller, rocker arm roller, etc. similarly to the roller followers. One end of the stud is so threaded as to facilitate mounting thereof. And the outer ring is guided in axial direction by the side plates which are press-fitted in the flange end of the stud and the stud itself.

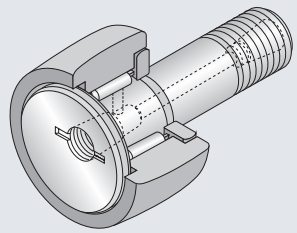
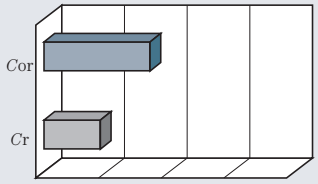
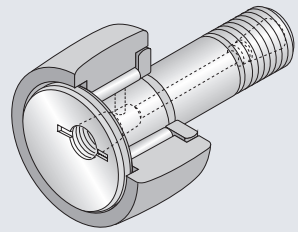
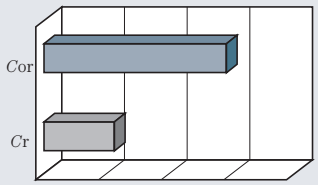
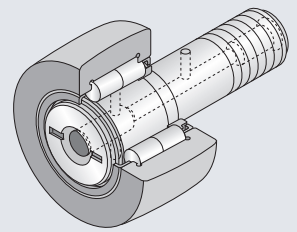
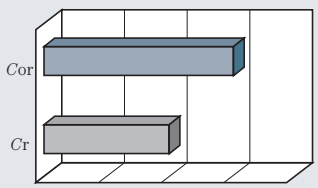
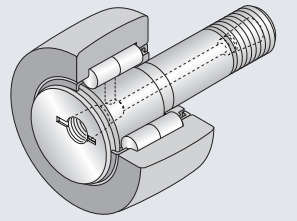
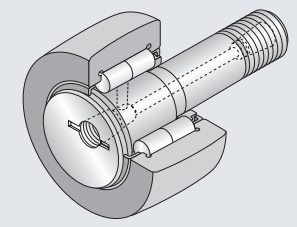
The outer ring has a thick wall and both of spherical outer profile and cylindrical outer surface (tail code: **X**) are available for it similarly to the outer ring of the roller follower.

Follower type	Applied shaft diameter (mm)	Load capacity	Composition of nominal bearing number
KR **CR** 	KR : ϕ3～ϕ30 CR : ϕ4.826～ϕ22.225	 **KR90**	**KR 12 T2 H / 3A** Tailed code T2: Resin cage H: With hexagon hole 3A: grease Dimension code Type code
KRT 	ϕ6～ϕ30		**KRT 32 X LL** Tail code LL: Seal Tailed code X: Cylinder outer diameter Dimension code Type code
KRV **CRV** 	KRV : ϕ3～ϕ30 CRV : ϕ4.826～ϕ63.5	 **KRV90**	**CRV 30 X LL** Tail code LL: Seal Tailed code X: Cylinder outer diameter Dimension code Type code
KRVT 	ϕ6～ϕ30		**KRVT 52 X LL** Tail code LL: Seal Tailed code X: Cylinder outer diameter Dimension code Type code

※Each listed load capacity is subject to the reference bearing diameter of ϕ30.

Follower components	Features
Outer ring outer diameter : ϕ12 With metric series cage Without seal Outer profile : spherical Stud head : with hexagon hole Cage : resin cage Grease : already filled	● Standard cage is of steel plate punched type. ● Polyamide resin cage (Tail code: T2) can be used, too. Allowable temperature: 120˚C Continuous operating temperature: 100˚C or less ● Suited to high speed. ● Due to much filled grease, these types can be used for a long time without additional greasing though actual operating conditions. ● The standard follower of Type KRT has no hexagon hole (tail code: H), but with a threaded plug with hexagon socket as standard. ● The follower with neither seal nor grease code as tail code is of grease non-filled type.
Outer ring outer diameter : ϕ32 With metric series cage Stud head : with recessed slot for screwdriver use and tapped hole Outer surface profile : cylindrical Seal : with Grease : already filled.	
Outer ring outer diameter : ϕ47.825 Inch series full-complement roller type Stud head : with recessed slot for screwdriver use Outer surface profile : cylindrical Seal : with Grease : already filled.	● Suited to high load. ● Lower allowable running speed than the type with cage. ● Grease replenishing interval must be shortened due to the small spacing volume. ● he standard follower of Type KRVT has no hexagon hole (tail code H), but with a threaded plug with hexagon socket as standard.
Outer ring outer diameter : ϕ32 With metric series cage Stud head : With recessed slot for screwdriver use and tapped hole Outer surface profile : cylindrical Seal : with Grease : already filled.	

Follower components	Features
Outer ring outer diameter : ϕ12 With metric series cage Without seal Outer profile : spherical Stud head : with hexagon hole Cage : resin cage Grease : already filled	● Standard cage is of steel plate punched type. ● Polyamide resin cage (Tail code: T2) can be used, too. Allowable temperature: 120˚C Continuous operating temperature: 100˚C or less ● Suited to high speed. ● Due to much filled grease, these types can be used for a long time without additional greasing though actual operating conditions. ● The standard follower of Type KRT has no hexagon hole (tail code: H), but with a threaded plug with hexagon socket as standard. ● The follower with neither seal nor grease code as tail code is of grease non-filled type.
Outer ring outer diameter : ϕ32 With metric series cage Stud head : with recessed slot for screwdriver use and tapped hole Outer surface profile : cylindrical Seal : with Grease : already filled.	
Outer ring outer diameter : ϕ47.825 Inch series full-complement roller type Stud head : with recessed slot for screwdriver use Outer surface profile : cylindrical Seal : with Grease : already filled.	● Suited to high load. ● Lower allowable running speed than the type with cage. ● Grease replenishing interval must be shortened due to the small spacing volume. ● he standard follower of Type KRVT has no hexagon hole (tail code H), but with a threaded plug with hexagon socket as standard.
Outer ring outer diameter : ϕ32 With metric series cage Stud head : With recessed slot for screwdriver use and tapped hole Outer surface profile : cylindrical Seal : with Grease : already filled.	

Follower type	Applied shaft diameter (mm)	Load capacity	Composition of nominal bearing number
KRU 	ϕ6～ϕ30	 **KRU90**	**KRU 32 LL** Tail code LL: seal Dimension code Type code
KRVU 	ϕ6～ϕ30	 **KRVU90**	**KRVU 62 X LL** Tail code LL: seal Tail code X: Cylinder outer diameter Dimension code Type code
NUKR 	ϕ12～ϕ64	 **NUKR90**	**NUKR 80 H** Tail code H: with hexagon socket Dimension code Type code
NUKRT 	ϕ12～ϕ64		**NUKRT 90** Dimension code Type code
NUKRU 	ϕ12～ϕ64		**NUKRU 140 X** Tail code X: Cylinder outer diameter Dimension code Dimension code

※Each listed load capacity is subject to the reference bearing diameter of ϕ30.

Follower components	Features
Outer ring outer diameter: ϕ32 Metric series stud with cage, eccentric type Stud head : with recessed slot for screwdriver use and tapped hole Outer profile : spherical Seal : with Grease : already filled	● Unlike Type KRT and KRVT, Type KRU and KRVU have the eccentric stud (eccentricity : 0.25 to 1.0mm) so as to enable to adjust positional variation of the stud mounting hole. ● The standard follower has no hexagon hole (tail code H), but with a threaded plug with hexagon socket as standard. ● Type KRU --without seal. In addition, the same follower with no grease code is of grease non-filled type.
Outer ring outer diameter: ϕ62 Metric series stud with cage, eccentric type Stud head : Metric series full-complement roller type stud, eccentric type Seal : with Outer profile : cylindrical Grease : already filled	
Outer ring outer diameter: ϕ80 Metric series double-row cylindrical roller type Shielded full-complement roller type Stud head : with hexagon socket Seal : with Outer profile : spherical Grease : already filled	● Highest rated load, best-suited to applications subjected to high load and shock load. ● The outer ring is guided in axial direction by the outer ring ribs and the end faces of cylindrical roller. ● Grease replenishing interval must be shortened due to the small spacing volume. ● Type NUKRU has the eccentric stud (eccentricity: 0.4 to 2.5mm) so as to enable to adjust positional variation of the stud mounting hole. ● The standard followers of Type NUKRT and NUKRU have no hexagon hole (tail code H), but with a threaded plug with hexagon socket as standard.
Outer ring outer diameter: ϕ90 Metric series double-row cylindrical roller type Shielded full-complement roller type Stud head : with recessed slot for screwdriver use and tapped hole Outer profile : spherical Grease : already filled	
Outer ring outer diameter: ϕ140 Metric series double-row cylindrical roller type Shielded full-complement roller type stud, eccentric type Stud head : with recessed slot for screwdriver use and tapped hole Outer profile : cylindrical Grease : already filled	

Bearing accuracy

The dimensional accuracy and profile accuracy of cylindrical roller outer diameter (D) and outer ring width (C), and the running accuracy of bearing assy are as shown in **Tables 4.3** and 4.4 of Section 4 "**Bearing Tolerances**" (page A-26) and the accuracy class of each conforms to JIS Class-0. The dimensional accuracy of spherical outer diameter (D) and stud diameter (d_1) are as shown in applicable Dimensions Table.

Bearing fit and radial internal clearance

Table 1 shows the recommended fitting tolerance for the stud mounting hole.

And **Table 2** shows the radial internal clearance.

Table 1 Recommended tolerance

Classification	Tolerance range class for mounting hole
Metric series	H7
Inch series	F7

Table 2 Radial internal clearance

Unit : μm

Nominal roller inscribed circle diameter F_w (mm)		Clearance							
		C2		CN (ordinary)		C3		C4	
over	incl.	min	max	min	max	min	max	min	max
3	6	0	10	3	17	15	30	20	40
6	10	0	12	5	20	15	30	25	45
10	18	0	15	5	25	15	35	30	55
18	30	0	20	10	30	20	40	40	65
30	50	0	25	10	40	25	55	50	80
50	80	0	30	15	50	30	65	60	100
80	100	0	35	20	55	35	75	70	115

Note) When shock load acts on, make the stud - hole clearance as less as possible in assembling.

Fitting relations

(1) Make the face height at the cam follower mount greater than "e" dimension given in applicable Dimension Table. (**Fig.1**)
Furthermore, chamfer the stud mounting hole at R as small as possible (around 0.5×45°) and bring the side faces of side face in precise contact.

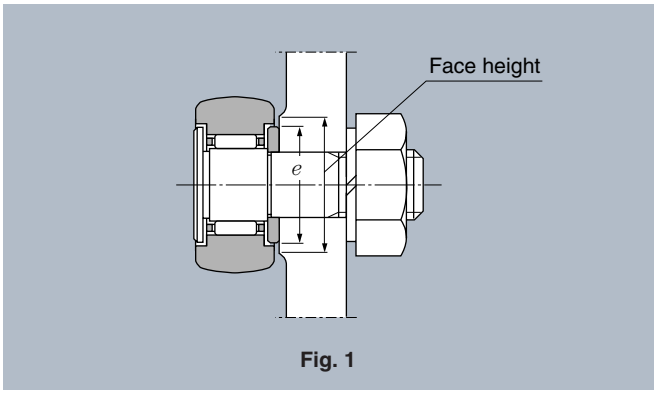


Fig. 1

(2) Don't hammer directly the cam follower rib. Doing so would cause breakdown and rotation failure of the rib.

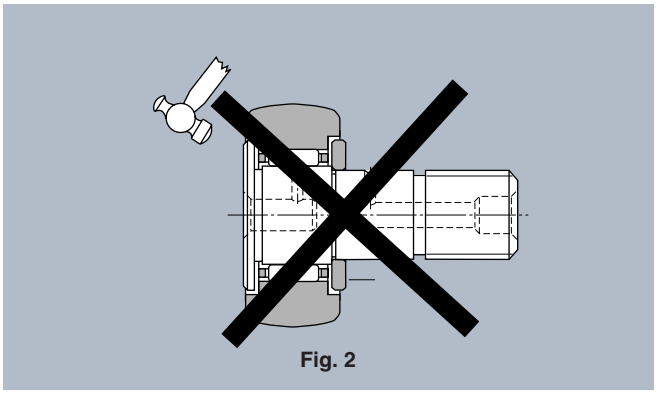


Fig. 2

(3) The oil hole position on the stud raceway surface is shown with **NTN** mark stamped on the stud rib surface. Mount the stud so the oil hole locates within the non-load area (non-load side). (**Fig. 3**) If the oil hole locates within the load area, it would cause shorter life of the follower.

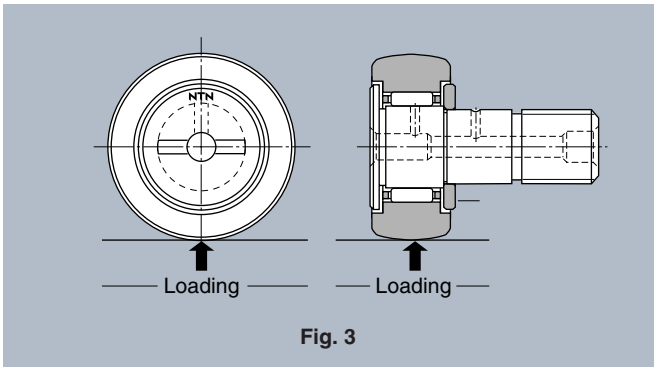
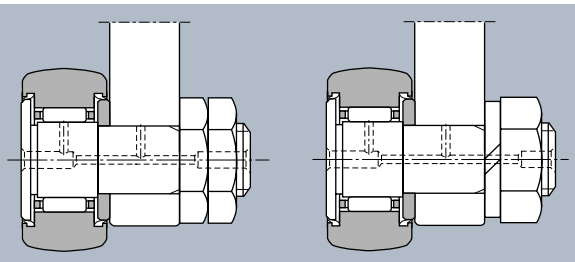


Fig. 3

(4) Particularly where loose of the mounting screw is forecast due to wide amplitude vibration during running, the mounting methods as illustrated in **Fig. 4** are available.



[a] Use JIS B 1181 Class-3 thin type nut as double nut for tightening.

[b] Use a specific washer with standard nut.

Fig. 4

(5) The stud is subjected to bending stress and tensile force arising from bearing load. Tighten the stud screws with tightening torque which does not exceed the torque value specified in applicable Dimension Table. **Too tight tightening torque could result in rupture of the threaded portion.**

(6) A hole is provided on the stud center at right angle to the shaft axial center, as illustrated in **Fig. 5**. Use this hole for locking or grease replenishment.

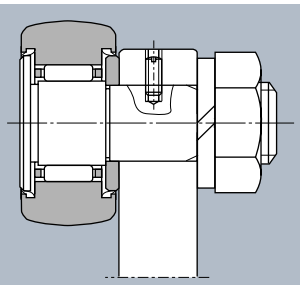


Fig. 5

(7) For mounting and adjusting the eccentric stud type cam follower, follow the sequence given hereunder.

1. Insert the stud in the mounting hole so that the **NTN** mark (showing oil hole position) locates in relative to load acting direction as illustrated in **Fig. 3**.
2. Turn the stud using the recessed slot for screwdriver on the stud head or the accessory threaded plug with hexagon socket, to adjust a clearance from the mating contact surface.
3. After adjustment, tighten the stud nut with specific tightening torque described in applicable Dimensions Table.

NTN cam follower is generally stud-mounted so that non-uniform load (bias load) could act on the bearing, inversely affected by fitting loose arising from continuous running.

So, good care must be exercised of fitting loose for stable running of an equipment.

Lubrication and how to feed and replenish grease into the follower

Lubrication

The cam follower type with synthetic rubber seal (with tail code LL) and the full-complement roller type are pre-filled up with lithium soap base grease so they can be used in the operating temperature range of -25 to +100˚C.

The type with cage and with no seal is not filled up with grease. If a grease pre-filled type is needed, feel free to contact **NTN**.

Further, **NTN** Bearings with Solid Drease using solid lubricant of less leakage is also available on request. Feel free to contact **NTN** for the detailed information.

The outer ring outer surface of bearing and the track surface must both be lubricated. Failure to lubricate could result earlier in damage of bearing.

How to feed and replenish grease

Feed and replenish grease through a grease nipple provided on the flange end face or threaded-side end face of the stud. In this case, plug the grease-feed hole or the tapped hole at non-greasing side (with a special-purposed plug or a threaded plug with hexagon socket.)

These grease nipple and plug are enclosed in each cam follower package. Screw-in them in specific position before mounting the cam follower.

Special grease nipple and that specified in JIS Standard as applicable are available for use. The dimensions thereof and the applicable bearing types are as specified in **Tables 3** and **4** respectively.

Furthermore, special press-fit type plug and threaded plug with hexagon socket are available for use. The dimensions thereof and the applicable bearing types are as specified in **Tables 5-1**, **-2** and **Table 6** respectively.

When using the special-purposed press-fit plug, press-fit it in the grease feed hole using a mandrel of applicable dimension shown in **Table 7**.

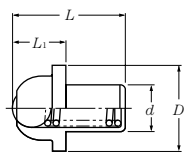
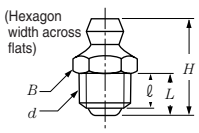


Table 3 Grease nipple dimension

Nominal nipple number	Dimension mm			
	d	D	L	L_1
NIP-B3	3	7.5	9	5.5
NIP-B4	4	7.5	10	5.5
NIP-B6	6	8	13	6
NIP-B8	8	10	16	7



Nominal nipple number	Dimension mm				
	d	H	ℓ	L	B
NIP-X30	M4×0.7	13.9	4	5	7
JIS A-M6F	M6×0.75	13.5	4	5.4	7
JIS A-PT1/8	PT1/8	20	8	9.5	10

Table 4 Grease nipple applied bearing types and grease nipple dimension codes

Nominal nipple number	Nipple applied bearing types									
	KR, KRV	KR··H, KRV··H	CRV	CRV··H	NUKR	NUKR··H	KRT, KRVT	NUKRT	KRU, KRVU	NUKRU
NIP-B3	—	—	8.8-1	—	—	—	—	—	—	—
NIP-B4	16～26	22～26	10～18	12～18	—	—	—	—	—	—
NIP-B6	30～40	30～40	20～44	20～44	30～40	30～40	—	—	—	—
NIP-B8	47～90	47～90	48～96	48～96	47～90	47～90	—	—	—	—
NIP-X30	—	—	—	—	—	—	16～26	—	16～26	—
JIS A-M6F	—	—	—	—	—	—	30～32	30	30～35	30～35
JIS A-PT1/8	—	—	—	—	100～180 (Threaded side)	100～180	35～90	35～180	40～90	40～180

Special-purposed plug

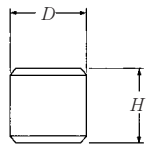


Table 5-1 Plug dimension

unit : mm

Nominal number	D	H
SEN 3	3	3
SEN 4	4	4
SEN 6	6	6
SEN 8	8	8

Threaded plug with hexagon socket

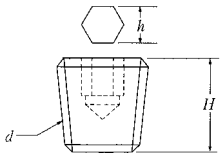


Table 5-2 Plug dimension

unit : mm

Nominal number	d	H	Width across flats h
M4X0.7X4 ℓ	M4X0.7	4	2
M6X0.75X6 ℓ	M6X0.75	6	3
PT1/8X7 ℓ	PT1/8	7	5

Table 6 Plug applied bearing types and plug dimension codes

Plug	Plug applied bearing types									
	KR, KRV	KR··H, KRV··H	CRV	CRV··H	NUKR	NUKR··H	KRT, KRVT	NUKRT	KRU, KRVU	NUKRU
SEN3	—	—	8.8-1	—	—	—	—	—	—	—
SEN4	16～26	22～26	10～18	12～18	—	—	—	—	—	—
SEN6	30～40	30～40	20～44	20～44	30～40	30～40	—	—	—	—
SEN8	47～90	47～90	48～96	48～96	47～180	47～90	—	—	—	—
M4X0.7X4 ℓ	—	—	—	—	—	—	16～26	—	16～26	—
M6X0.75X6 ℓ	—	—	—	—	—	—	30～32	30	30～35	30～35
PT1/8X7 ℓ	—	—	—	—	—	100～180	35～90	35～180	40～90	40～180

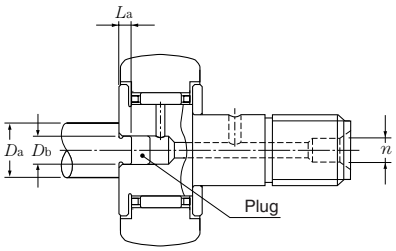


Table 7 Mandrels used for plug press-fitting

unit : mm

Grease feed hole dia. n	Mandrel dimension D_a	$D_b{}_{-0.1}^{0}$	$L_a{}_{-0.1}^{0}$
3	8	2.8	1.5
4	10	3.8	1.5
6	12	5.8	1.5
8	15	7.8	2.5

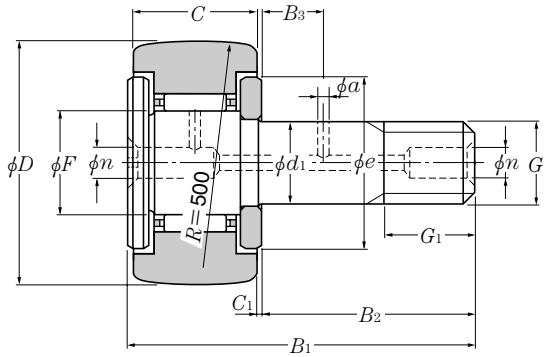
Track load capacity

Maximum radial load which is statically allowable on the track roller to track contact surface is referred to as "track load capacity", and the track load capacity subject to track surface hardness HRC40 is as described in applicable Dimensions Table.

Where the track surface hardness is other than HRD40, adjust the track load capacity using the adjustment factor shown in **Table 11.1**.

Metric series

Type KR
Type KR··LL
Type KRV
Type KRV··LL



Type KR
(With cage)

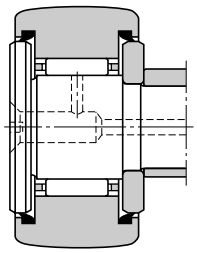
D 10～52mm

Boundary dimensions

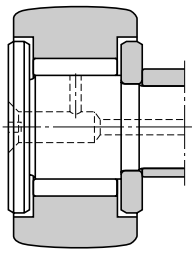
mm

D $^{0}_{-0.050}$	C	d_1	F	B_1	B_2	G	G_1	C_1	n	B_3	a	e	h
10	7	3 $^{0}_{-0.010}$	4.0	17	9	M 3×0.5	5	0.5	—	—	—	7	2.5[3]
	7	3 $^{0}_{-0.010}$	4.0	17	9	M 3×0.5	5	0.5	—	—	—	7	2.5[3]
12	8	4 $^{0}_{-0.012}$	4.8	20	11	M 4×0.7	6	0.5	—	—	—	8.5	2.5[3]
	8	4 $^{0}_{-0.012}$	4.8	20	11	M 4×0.7	6	0.5	—	—	—	8.5	2.5[3]
13	9	5 $^{0}_{-0.012}$	5.75	23	13	M 5×0.8	7.5	0.5	—	—	—	9.5	3[3]
	9	5 $^{0}_{-0.012}$	5.75	23	13	M 5×0.8	7.5	0.5	—	—	—	9.5	3[3]
16	11	6 $^{0}_{-0.012}$	8	28	16	M 6×1	8	0.6	4[2]	—	—	12	3
	11	6 $^{0}_{-0.012}$	8	28	16	M 6×1	8	0.6	4[2]	—	—	12	3
19	11	8 $^{0}_{-0.015}$	10	32	20	M 8×1.25	10	0.6	4[2]	—	—	14	4
	11	8 $^{0}_{-0.015}$	10	32	20	M 8×1.25	10	0.6	4[2]	—	—	14	4
22	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	4	—	—	17	4
	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	4	—	—	17	4
26	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	4	—	—	17	4
	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	4	—	—	17	4
30	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	6	6	3	23	6
	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	6	6	3	23	6
32	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	6	6	3	23	6
	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	6	6	3	23	6
35	18	16 $^{0}_{-0.018}$	18	52	32.5	M16×1.5	17	0.8	6	8	3	27	6
	18	16 $^{0}_{-0.018}$	18	52	32.5	M16×1.5	17	0.8	6	8	3	27	6
40	20	18 $^{0}_{-0.018}$	22	58	36.5	M18×1.5	19	0.8	6	8	3	32	6
	20	18 $^{0}_{-0.018}$	22	58	36.5	M18×1.5	19	0.8	6	8	3	32	6
47	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	8	9	4	37	8
	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	8	9	4	37	8
52	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	8	9	4	37	8
	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	8	9	4	37	8

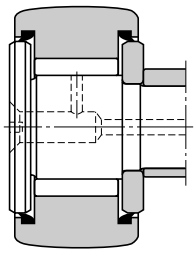
Note 1) For bearing with cylindrical outer surface, the nominal number is followed by "**X**" code. In this case, the cylindrical bearing is manufactured with the tolerance for outer ring outer diameter D which conforms to JIS Class-0. **Ex. KR22XLL**
2) Grease replenishing hole is provided at front side only (on left side-face in the above sketch). However, in the case with "hexagon hole" no grease replenishing hole is provided.



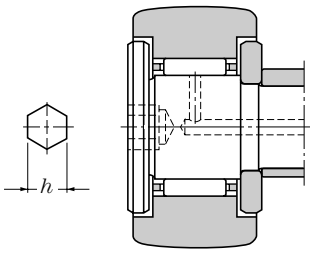
Type KR··LL
(Sealed type with cage)



Type KRV
(Full complement type)



Type KRV··LL
(Full complement type with seal)



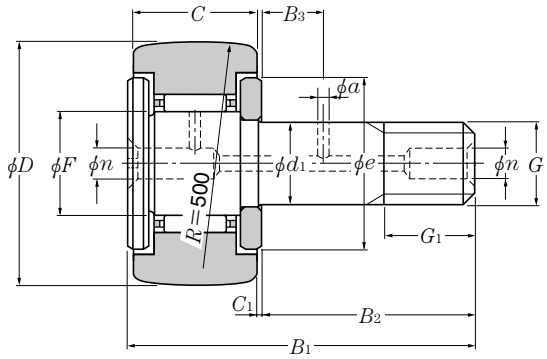
Type KR··H
(With hexagonal wrench socket)

Basic load ratings dynamic N C_r	Basic load ratings static N C_{or}	Basic load ratings dynamic kgf C_r	Basic load ratings static kgf C_{or}	Load ratings of truck N Spherical outer ring	Load ratings of truck N Cylindrical outer ring	Load ratings of truck kgf Spherical outer ring	Load ratings of truck kgf Cylindrical outer ring	Limiting speeds r/min grease	Limiting speeds r/min oil	Maximum tightening torque N·m	Maximum tightening torque kgf·m	Bearing numbers Type NR [1] Type KRV	Bearing numbers Type NR··LL [1] Type KRV··LL	Mass (approx.) kg Type KR	Mass (approx.) kg Type KRV
1 640	1 270	168	130	560	1 360	57	139	※27 000	※40 000	0.5	0.05	KR10T2H/3A	KR10T2LLH	0.005	—
2 500	2 610	255	267	560	1 360	57	139	※25 000	※32 000	0.5	0.05	KRV10H	KRV10LLH	—	0.005
2 170	1 690	221	172	725	1 790	74	183	※25 000	※36 000	1	0.1	KR12T2H/3A	KR12T2LLH	0.008	—
3 500	3 800	360	385	725	1 790	74	183	※20 000	※27 000	1	0.1	KRV12H	KRV12LLH	—	0.008
2 650	2 260	270	231	805	2 220	82	226	※23 000	※33 000	2	0.2	KR13T2H/3A	KR13T2LLH	0.010	—
4 500	5 350	455	545	805	2 220	82	226	※17 000	※22 000	2	0.2	KRV13H	KRV13LLH	—	0.011
4 050	4 200	415	430	1 080	3 400	110	350	※19 000	※25 000	3	0.3	KR16	KR16LL	0.019	—
6 500	9 350	665	955	1 080	3 400	110	350	※13 000	※16 000	3	0.3	KRV16	KRV16LL	—	0.020
4 750	5 400	480	555	1 380	4 050	141	415	※15 000	※20 000	8	0.8	KR19	KR19LL	0.031	—
7 450	11 700	760	1 190	1 380	4 050	141	415	10 000	※13 000	8	0.8	KRV19	KRV19LL	—	0.032
5 300	6 650	540	680	1 690	5 150	172	525	※12 000	※16 000	14	1.4	KR22	KR22LL	0.046	—
8 200	14 000	840	1 420	1 690	5 150	172	525	8 500	※11 000	14	1.4	KRV22	KRV22LL	—	0.047
5 300	6 650	540	680	2 120	6 100	216	620	※12 000	※16 000	14	1.4	KR26	KR26LL	0.059	—
8 200	14 000	840	1 420	2 120	6 100	216	620	8 500	※11 000	14	1.4	KRV26	KRV26LL	—	0.061
7 850	9 650	800	985	2 620	7 700	267	785	10 000	※13 000	20	2	KR30	KR30LL	0.087	—
12 000	20 300	1 230	2 070	2 620	7 700	267	785	6 500	8 500	20	2	KRV30	KRV30LL	—	0.089
7 850	9 650	800	985	2 860	8 200	291	835	10 000	※13 000	20	2	KR32	KR32LL	0.097	—
12 000	20 300	1 230	2 070	2 860	8 200	291	835	6 500	8 500	20	2	KRV32	KRV32LL	—	0.100
12 200	17 900	1 240	1 830	3 200	11 900	325	1 220	8 000	※11 000	52	5.3	KR35	KR35LL	0.169	—
17 600	34 000	1 790	3 500	3 200	11 900	325	1 220	5 500	7 000	52	5.3	KRV35	KRV35LL	—	0.172
14 000	22 800	1 430	2 330	3 850	14 500	390	1 480	7 000	9 000	76	7.8	KR40	KR40LL	0.248	—
19 400	42 000	1 980	4 250	3 850	14 500	390	1 480	4 500	6 000	76	7.8	KRV40	KRV40LL	—	0.252
20 700	33 500	2 110	3 450	4 700	21 000	480	2 150	6 000	8 000	98	10	KR47	KR47LL	0.386	—
28 800	61 000	2 940	6 250	4 700	21 000	480	2 150	4 000	5 000	98	10	KRV47	KRV47LL	—	0.390
20 700	33 500	2 110	3 450	5 550	23 300	565	2 370	6 000	8 000	98	10	KR52	KR52LL	0.461	—
28 800	61 000	2 940	6 250	5 550	23 300	565	2 370	4 000	5 000	98	10	KRV52	KRV52LL	—	0.465

Note 3) Cam follower with hexagon hole is standard.
Remarks: For the cam follower with ※marked seal, the allowable running speed is approximately 10 000 r/min.

Metric series

Type KR
Type KR··LL
Type KRV
Type KRV··LL



Type KR
(With cage)

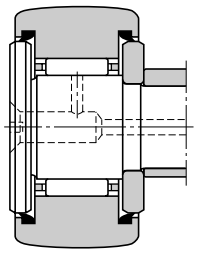
D 62～90mm

Boundary dimensions

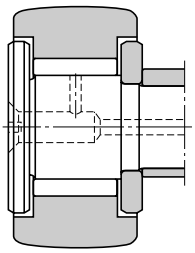
mm

D $^{0}_{-0.050}$	C	d_1	F	B_1	B_2	G	G_1	C_1	n	B_3	a	e	h
62	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	8	11	4	44	8
	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	8	11	4	44	8
72	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	8	11	4	44	8
	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	8	11	4	44	8
80	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	8	15	4	53	8
	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	8	15	4	53	8
85	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	8	15	4	53	8
90	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	8	15	4	53	8
	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	8	15	4	53	8

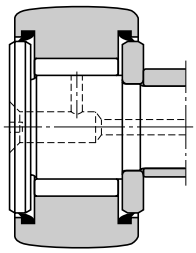
Note 1) For bearing with cylindrical outer surface, the nominal number is followed by "**X**" code. In this case, the cylindrical bearing is manufactured with the tolerance for outer ring outer diameter D which conforms to JIS Class-0. **Ex. KR62XLL**



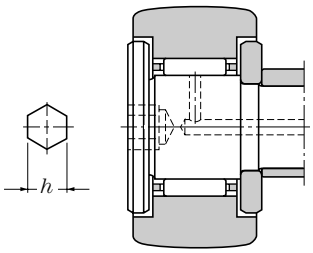
Type KR··LL
(Sealed type with cage)



Type KRV
(Full complement type)



Type KRV··LL
(Full complement type with seal)

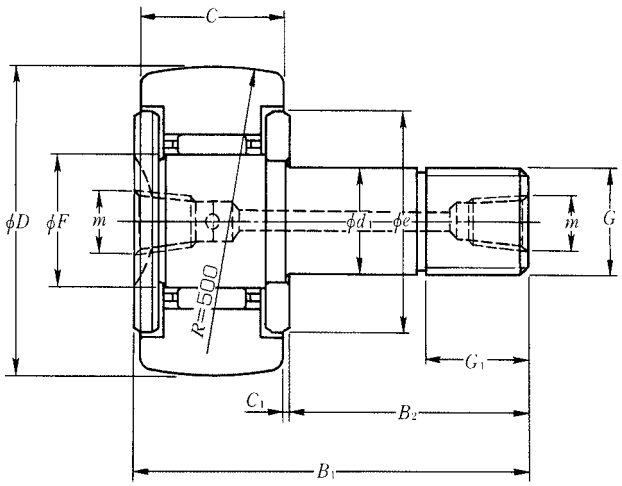


Type KR··H
(With hexagonal wrench socket)

Basic load ratings dynamic N		Basic load ratings static N		Load ratings of truck N				Limiting speeds r /min		Maximum tightening torque		Bearing numbers		Mass (approx.) kg	
C_r	C_{or}	C_r	C_{or}	Spherical outer ring	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	grease	oil	N·m	kgf·m	Type NR [1] Type KRV	Type NR··LL[1] Type KRV··LL	Type KR	Type KRV
28 900	55 000	2 950	5 600	6 950	34 500	710	3 500	5 000	6 500	178	18	**KR62**	**KR62LL**	0.790	—
39 500	98 500	4 000	10 000	6 950	34 500	710	3 500	3 300	4 500	178	18	**KRV62**	**KRV62LL**	—	0.800
28 900	55 000	2 950	5 600	8 050	38 500	820	3 900	5 000	6 500	178	18	**KR72**	**KR72LL**	1.04	—
39 500	98 500	4 000	10 000	8 050	38 500	820	3 900	3 300	4 500	178	18	**KRV72**	**KRV72LL**	—	1.05
45 000	88 500	4 600	9 050	9 800	53 000	1 000	5 400	40 00	5 500	360	37	**KR80**	**KR80LL**	1.55	—
58 000	147 000	5 900	15 000	9 800	53 000	1 000	5 400	2 600	3 500	360	37	**KRV80**	**KRV80LL**	—	1.56
45 000	88 500	4 600	9 050	10 400	56 000	1 060	5 750	4 000	5 500	360	37	**KR85**	**KR85LL**	1.74	—
45 000	88 500	4 600	9 050	11 400	59 000	1 160	6 100	4 000	5 500	360	37	**KR90**	**KR90LL**	1.95	—
58 000	147 000	5 900	15 000	11 400	59 000	1 160	6 100	2 600	3 500	360	37	**KRV90**	**KRV90LL**	—	1.97

Metric series

Type KRT
Type KRT··X
Type KRT··LL
Type KRT··XLL



Type KRT
(With cage)

D 16～90mm

Boundary dimensions mm											Basic load ratings dynamic N	static N	dynamic kgf	static kgf
D 1)	C	d_1	F	B_1	B_2	G	G_1	C_1	e	m	C_r	C_{or}	C_r	C_{or}
16	11	$6\ ^{0}_{-0.012}$	8	28	16	M 6×1	8	0.6	12	M4×0.7 2)	4 050	4 200	415	430
19	11	$8\ ^{0}_{-0.015}$	10	32	20	M 8×1.25	10	0.6	14	M4×0.7 2)	4 750	5 400	480	555
22	12	$10\ ^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	5 300	6 650	540	680
26	12	$10\ ^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	5 300	6 650	540	680
30	14	$12\ ^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	7 850	9 650	800	985
32	14	$12\ ^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	7 850	9 650	800	985
35	18	$16\ ^{0}_{-0.018}$	18	52	32.5	M16×1.5	17	0.8	27	PT1/8	12 200	17 900	1 240	1 830
40	20	$18\ ^{0}_{-0.018}$	22	58	36.5	M18×1.5	19	0.8	32	PT1/8	14 000	22 800	1 430	2 330
47	24	$20\ ^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	20 700	33 500	2 110	3 450
52	24	$20\ ^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	20 700	33 500	2 110	3 450
62	29	$24\ ^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	28 900	55 000	2 950	5 600
72	29	$24\ ^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	28 900	55 000	2 950	5 600
80	35	$30\ ^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	45 000	88 500	4 600	9 050
85	35	$30\ ^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	45 000	88 500	4 600	9 050
90	35	$30\ ^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	45 000	88 500	4 600	9 050

Note 1) The dimensional tolerances for the outer ring outer diameter *D* of **Type KRT··X** and **KRT··XLL** with cylindrical outer surface conform to JIS Class -0.
2) A tapped hole is provided at front side only (on left end-face in the above sketch).